MILITARY SPECIFICATION

POWDER MATERIALS

FOR FLAME AND PLASMA SPRAYED COATINGS

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes three (3) material categories for use in constructing thermal spray coatings, consisting of metallic and/or non-metallic components. These powdered materials may be suitable for application using either plasma arc or combustion gas (flame spray) techniques.

1.2 Classification. Thermal spray powders covered in this specification shall be according to the following Types (chemical composition) and Classes (particle size distribution).

1.2.1 Types.

Type I - Metallics
Type II - Non-Metallics
Type III - Mixtures of either Type I and Type I, Type I and Type II or Type II and Type II materials.

1.2.2 Classes.

Class 1 - Fine powders, predominately less than 45 microns (-325 mesh).
Class 2 - Coarse powders, predominately greater than 45 microns (+325 mesh).

1.2.3 Compositions - As defined in individual detail specification sheets.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3439
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) ans supplement thereto cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL

PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals

STANDARDS
FEDERAL

FED-STD-313	-	Material Safety Data Sheets, Preparation and Submission of

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage

(See supplement 1 for list of associated specifications)

(Unless otherwise indicated, copies of federal and military specifications ans standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 214	-	Standard Method of Test for Sieve Analysis of Granular Metal Powder
ASTM B 215	-	Sampling Finished Lots of Metal Powders
ASTM C 117	-	Test Method for Materials Finer Than 75 um (no. 200) Sieve in Mineral Aggregates by Washing.
ASTM C 633	-	Adhesion of Cohesive Strength of Flame Sprayed Coatings

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ASTM E 11 - Wire Cloth Sieves for Testing Purposes.

ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

ASTM E 354 - Metallic Chemical Analysis of High Temperature, Electrical, and Other Similar Iron, Nickel, and Cobalt Alloys

ASTM E 384 - Standard Test Method for Microhardness

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable detail specifications. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern.

3.2 First article. First article inspection may consist of a preproduction sample or a first article sample selected from the first production lot of the contract or purchase order. It shall be inspected in accordance with 4.4.(see 6.3).

3.3 Chemical and physical properties. The chemical and physical properties shall conform to 3.3.1 through 3.3.4, as specified in the detail specification sheet, when tested as specified in 4.6.

3.3.1 Visual appearance. All thermal spray powders shall be of uniform shape and color; clean; dry, non-packing; and free of contaminants as determined in 4.6.1.

3.3.2 Chemical composition. The chemical composition shall be as specified in the applicable detail specification sheet and determined in accordance with 4.6.2. The formulation and source of raw material shall be maintained and recorded in accordance with 4.5.1.

3.3.3 Particle size/distribution analysis. The powder particle size and distribution shall be as specified in the applicable specification sheet and determined in accordance with 4.6.3.

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3.3.4 Sprayability. The material's ability to flow in a satisfactory manner and construct an acceptable thermal spray deposit shall be demonstrated. Acceptance criteria shall be as specified herein and in the applicable specification sheet when determined in accordance with 4.6.4.

3.3.4.1 Macroscopic examination. The as-sprayed coating when examined at 10 to 20x magnification shall have a uniform appearance. Surface defects shall be limited to small nodules not to exceed 1.14mm (0.045 in) in diameter and shall not exceed 0.38mm (0.015 in) above the surrounding sprayed surface. The coating shall not contain any:

(a) Blisters
(b) Cracks
(c) Chips or loosely-adhering particles
(d) Pits exposing the undercoat or substrate

3.3.4.2 Metallographic examination. Coatings when prepared using standard metallographic laboratory methods for thermal spray coatings shall show microstructures containing the specific characteristics for the coating composition under evaluation including interface condition, cracks, porosity, oxides and oxide stringers, globular particles, unreacted particles and uniformity of constituents. Coating microstructures shall be free from inclusions and contamination at the coating-base metal interface and from cracks, massive porosity and excessive oxides to the extent specified in the appropriate specification sheet.

3.3.4.2.1 Interface condition. Up to 20% of the interface of a cross-section not less than 1.91mm (0.75 in) in length containing grit, fine contamination or oxides with no separation is acceptable.

3.3.4.2.2 Cracks. Coatings shall be free from cracks. Microcracks that are a normal characteristic of a particular coating composition are not rejectable.

3.3.4.2.3 Porosity. Refer to the applicable detail specification sheet for limits on coating porosity. Porosity in excess of that specified in the detail specification sheet or massive porosity is cause for rejection.

3.3.4.2.4 Oxides, globular particles, unreacted particles. Refer to the coating microstructure requirements on the applicable detail specification sheet.

3.3.4.2.5 Uniformity of constituents. Unless otherwise specified by the detail specification sheet, coatings shall show a uniform distribution of microstructural constituents.

3.3.5 Deposit efficiency. Unless otherwise specified, the deposit efficiency shall be as specified in the applicable detail specification sheet when tested in accordance with 4.6.6.

3.6 Material safety data sheets. The contracting activity and the Naval Air Engineering Center (NAEC) shall be provided with a Material Safety Data Sheet (MSDS) at the time, of the contract award. The MSDS is form OSHA-20 and is found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification. (see 6.5)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the documents referenced in 4.5.

4.4 First article inspection. First article inspection shall be conducted at a laboratory satisfactory to Naval Air Engineering Center, Systems Engineering and Standardization Department, Materials Branch (Code 5314), Lakehurst NJ 08733-5100. First article inspection shall consist of the examinations and tests specified in 4.6. Reports of first article tests shall be provided to the contracting officer and to the Naval Air Engineering Center.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each powder lot prior to being offered for acceptance under contract. The quality conformance inspection shall consist of the examinations and tests specified in Table I. The vendor, contractor, or powder manufacturer shall provide test results with each powder lot, demonstrating the powder's conformance to the applicable specification sheet, and an equipment parameter sheet as shown in Figures 1 and 2. The spray parameters in accordance with Figure 2, which are required to produce the coating properties of the applicable detail specification shall be provided, as a minimum for the commercial equipment used to establish the properties. Failure of the powder samples to pass any of the examinations or tests shall be cause for rejection of the lot. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Group A		
Visual	3.3.1	4.6.1
Filled containers	5.1	4.6.7
Group B		
Chemical composition	3.3.2	4.6.2
Particle size/distribution	3.3.3	4.6.3
Group C		
Sprayability	3.3.4	4.6.4 4.6.5
Deposit efficiency	3.3.5	4.6.6

4.5.1 Inspection records. The contractor shall keep inspection records complete and available to the procuring activity for not less than three years from the date the powder is shipped.

4.5.2 Sampling.

4.5.2.1 Powders. Test samples for inspection of powders shall be taken from inspection units selected at random from each lot or batch in accordance with ASTM B 215. Sufficient powder shall be taken from each inspection unit to perform not less than two (2) evaluations for each test.

4.5.2.2 Containers. A random sample of filled containers shall be selected from each lot of powder and tested per 4.6.7.

4.6 Test methods.

4.6.1 Visual inspection. The powder shall be visually inspected at 1x magnification to determine conformance to 3.3.1 relative to color, cleanliness, dryness and non-packing attributes. Inspection for contaminants and uniformity shall be conducted at magnifications up to 40x.

4.6.2 Chemical composition. Chemical composition shall be determined by emission spectrochemical analysis, by wet chemical analysis in accordance with ASTM E 354 or ASTM E 363, or by other methods approved by the cognizant engineering activity.

4.6.3 Particle size. Powders shall be tested in accordance with ASTM B 214, ASTM C 117, or other method acceptable to the cognizant engineering activity.

4.6.4 Sprayability.

4.6.4.1 Material sprayability. A test panel nominally constructed of an annealed ferrous or non-ferrous metal, or as prescribed by contract or purchase order shall be suitably prepared to receive the thermal spray deposit. The test panel shall be not less than 25.4mm (1.0 in) x 76.2mm (3.0 in) x 2.54mm (0.10 in). Spray application shall utilize commercial, readily available thermal spray equipment (see 4.3 and 4.4).

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4.6.4.1.1 Macroscopic examination. The as-sprayed coating when examined at 10 to 20x magnification shall meet the requirements of 3.3.4.1 and the applicable detail specification sheet.

4.6.4.1.2 Microscopic examination. When examined at 250x the coating shall meet the requirements of 3.3.4.2 and the applicable detail specification sheet.

4.6.4.1.3 Bond strength. Coating adhesive/cohesive strength shall be determined in accordance with ASTM C 633. A minimum of three (3) specimens per spray run shall be required. All test results shall be reported.

4.6.4.1.4 Hardness testing. When macrohardness and/or microhardness limits are specified on the detail specification sheet, testing shall be in accordance with this paragraph. The macrohardness of the sprayed coating shall be conducted in accordance with ASTM E 18. A minimum of five (5) readings per sample are required. Hardness shall be taken on a smooth ground surface; minimum coating thickness for the 15N and 15T scales is 0.4mm (0.015 in) and 19.1mm (0.075 in) for the 15Y scale. Microhardness, shall be determined per ASTM E 384, using the Vicker's (pyramidal) diamond indentor. A minimum of ten (10) readings per specimen are required.

4.6.4.1.5 Non-established requirements. When requirements are not specified in the detail specification sheet, the test data shall be submitted to the procuring activity for evaluation. The submitted test data shall include porosity content, oxide content, bond strength and photomicrographs of representative microstructure of the coating.

4.6.5 Feedability. Feedability shall be determined in accordance with ASTM B 213. Visual observation of the passage/exit of the thermal spray powder from the powder injector port and into the gun's heat zone shall show its passage to be a steady, uninterrupted, constant stream without noticible hesitation. This shall be conducted for a ten (10) minute, continuous period with the thermal spray device ignited.

4.6.6 Deposit efficiency. The deposit efficiency shall be determined by spraying a test panel not less than 50.8mm (2 inches) by 101.6mm (4 inches) and determining percent by weight of material deposited compared with that sprayed. The deposit efficiency shall not be less than the limit defined by the applicable detail specification sheet.

4.6.7 Examination of filled containers. Each of the filled containers selected in accordance with 4.5.2 shall be examined for defects of container and closure, for evidence of leakage and for unsatisfactory markings to determine conformance with 5.1.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with the requirements of PPP-C-186. The containers shall be moisture proof and unbreakable and lids shall be fully sealed.

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5.1.1 Quantities. Unless otherwise specified by the procuring activity powder shall be packaged in 2kg individual unit containers. Each carton shall contain no more than twenty four (24) kg of material.

5.2 Coating instruction sheet. Each unit container shall contain as a minimum, an instruction sheet with the spray parameters as required by the applicable specification sheet in the manner detailed in Figure 2. The method of application shall be includes on the instruction sheet.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The powders covered by this specification are intended for use with thermal spray equipment and are to be applied as coatings to metallic or nonmetallic surfaces to rebuild worn or undersize areas, improve wear resistance, provide erosion and corrosion protection, to provide clearance control coatings, and for other uses.

6.2 Acquisition requirements. Acquistion documents must specify the following:

a. Title, number (including revision letter), and date of this specification and the applicable specification sheet.

b. Issue of DODISS to be cited in the solicition, and if required, the specific issue of individual documents referenced (see 2.2).

c. Type, Class, and Composition (see 1.2).

d. Manufacturer's product designation.

e. Quantity.

f. Size of container, if different than 5.1.1.

g. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID is tailored to reflect the requirements of the specific acquisition. To ensure correct contractural application of the data requirements, a Contract Data Requirements list (DD Form 1423) must be prepared to obtain the Data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID number	DID Title	Suggested Tailoring
3.2 & 4.4	DI-T-4902	First Article Inspection Report	Except for paragraprh 7

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The above DID was cleared as of the date of this specification. The current isssue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer shall provide specific guidance to offerers whether the powder should be a preproduction sample or a first artilce sample selected from the first production lot (see 3.2) and the number of samples to be tested as specified in 4.5.2. The contracting officer shall also include specific instructions in the acquisition documents regarding arrangements for examinations, approvals of first article test results by Naval Air Engineering Center, SESD(Code 5321), Lakehurst NJ 08733-5100 and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a powder which has been previously acquired or tested by the Government, and that bidders offering such powder, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solititation.

6.5 Definition of terms.

6.5.1 Flame spraying - A thermal spraying process in which an oxyfuel gas flame is the source of heat for melting the surfacing material. Compressed gas may or may not be used for atomizing and propelling the powder material to the substrate.

6.5.2 Plasma spraying - A thermal spraying process in which a nontransferred arc is utilized as the source of heat that ionizes a gas which melts and propels the coating material to the workpiece.

6.5.3 Batch or Lot - A batch or lot of powder shall be that product produced from a single uniform blend of powder produced at one time, from one or more furnace charges and presented for inspection at one time.

6.5.4 Capability - The words "shall be capable of" or "capability test" indicate characteristics or properties required in the product but for which testing of each lot is not required. However, if such testing is performed by the purchaser, material not conforming to the requirements shall be subject to rejection.

6.5.5 Feedability - That powder characteristic that enables it to flow in an uninterrupted, steady stream and behaving as a fluid liquid such that when fed through a suitable feed system it will not clog nor pulsate.

6.5.6 Sprayability - The powder characteristic which results in tightly adherent thermal spray coatings which are smooth, uniform and free from lumps and cracks, when sprayed in accordance with specified process parameters.

6.5.7 Contaminants - Anything not specifically included in the stated chemical composition.

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6.6 Similiar specification. MIL-P-83348A, Powder, PLasma Spray is listed for information only and shall not be construed as an acceptable alternative.

6.7 Material safety data sheet. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.8 Subject term (keyword) listing.

Coatings, Metallic
Coatings, Non-metallic
Coatings, Flame sprayed
Coatings, Plasma sprayed
Equipment, spray
Flame spraying
Metal spraying
Plasma
Plasma spraying
Powders
Thermal spraying

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH
DLA - GS

Preparing Activity
Navy - AS
(Project No. 3439-0663)

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SUPPLIER _____

POWDER TYPE _____ LOT NUMBER _____

EQUIPMENT TYPE _____

MODEL _____ S/N _____

_____

TEST RESULTS
FIRST SET OF SPECIMENS

BOND STRENGTH RESULTS _____ HARDNESS RESULTS _____

MICROSTRUCTURE EVALUATION:

CRACKS _____ POROSITY _____ OXIDES _____

GLOBULAR PARTICLES _____

UNREACTED PARTICLES _____

UNIFORMITY OF CONSTITUENTS _____

OTHER _____

SECOND SET OF SPECIMENS

BOND STRENGTH RESULTS _____ AVG. HARDENESS RESULTS _____

MICROSTRUCTURE EVALUATION:

CRACKS _____ POROSITY _____ OXIDES _____

GLOBULAR PARTICLES _____

UNREACTED PARTICLES _____

UNIFORMITY OF CONSTITUENTS _____

OTHER _____

APPROVAL STATUS: (ACCEPT, REJECT) _____

_____
Certifying Agent

Figure 1. Test Results.

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COMPANY _____

ADDRESS _____

MATERIAL SPECIFICATION _____ LOT# _____

SURFACE PREPARATION: GRIT _____ TYPE _____ SIZE _____ PRESSURE _____
SUBSTRATE MATERIAL USED: _____
PROCESS (CHECK ONE) PLASMA _____ FLAME SPRAY _____

EQUIPMENT MFG: _____
MODEL _____ S/N _____

POWDER FEEDER _____ TYPE _____ MFG _____ MODEL _____ S/N _____

OPERATING CONDITIONS:

Plasma	Combustion Gas
Working Gas _____	Nozzle _____
Nozzle _____	Metering Valve or Wheel _____
Cathode _____	Wheel RPM _____
Cathode Spacing _____	Gases: Type / PSI / Flow
Powder Port _____	____ ___ ____
Gas: Primary Type: _____ Purity: _____	____ ___ ____
PSI _____	____ ___ ____
Flow _____	____ ___ ____
Secondary Type: _____ Purity: _____	____ ___ ____
PSI _____	____ ___ ____
Flow _____	____ ___ ____
Carrier _____	Carrier Plate _____
Flow _____	
Voltage Current _____	Spray Rate _____
Feed Rate _____	Spray Distance _____
Spray Distance _____	Other _____
Other _____	

Figure 2. Equipment Parameter Report.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-85856	POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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